

ELORA RESEARCH STATION, ON, Canada

WMO: 713520

Lat: 43.6420N

Lon: 80.4117W

Elev: 1235

StdP: 14.05

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.0	(c) -0.7	(d) -11.0	(e) 3.2	(f) -3.9	(g) -5.3	(h) 4.3	(i) 0.2	(j) 27.9	(k) 23.9	(l) 25.3	(m) 21.8	(n) 6.4	(o) 320	(p) 0.592

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	84.9	71.3	82.1	70.3	79.5	68.7	75.0	80.6	72.8	79.0	70.8	76.9	10.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	129.8	78.1	70.8	118.9	75.9	68.8	110.9	74.2	39.4	80.8	37.3	79.4	35.5	77.1	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 24.5	(b) 21.5	(c) 19.2	(d) DB	(e) -12.6	(f) 89.0	(g) 6.2	(h) 2.6	(i) -17.1	(j) 90.8	(k) -20.8	(l) 92.3	(m) -24.3	(n) 93.8	(o) -28.8	(p) 95.7	(q) 87.1
(a) 24.5	(b) 21.5	(c) 19.2	(d) WB	(e) -13.2	(f) 78.1	(g) 6.1	(h) 3.5	(i) -17.6	(j) 80.6	(k) -21.2	(l) 82.6	(m) -24.7	(n) 84.5	(o) -29.1	(p) 87.1	(q) 87.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	19.2	20.2	29.2	41.9	54.0	63.6	66.9	65.4	59.0	47.4	35.8	26.0
	DBStd	19.10	11.48	10.68	11.07	8.73	8.40	6.65	5.37	5.19	7.07	8.06	8.79	9.37
	HDD50	4092	954	833	651	270	52	1	0	0	8	150	429	744
	HDD65	7871	1419	1253	1110	694	356	103	42	59	205	546	875	1209
	CDD50	1972	0	0	6	26	176	409	524	478	279	70	4	0
	CDD65	275	0	0	0	0	15	61	101	72	25	1	0	0
	CDH74	2391	0	0	1	8	171	545	852	558	239	17	0	0
	CDH80	484	0	0	0	0	28	132	183	97	43	1	0	0
(14) Wind	WSAvg	8.9	10.6	10.4	10.1	10.4	9.0	7.7	6.7	6.2	6.9	8.5	9.6	10.4
(15) Precipitation	PrecAvg	35.0	2.9	2.3	2.2	3.0	3.3	3.3	3.4	2.9	3.0	3.1	3.1	2.7
	PrecMax	44.7	4.8	4.8	5.4	5.1	6.2	7.2	6.1	6.4	6.0	5.4	6.0	4.9
	PrecMin	25.5	1.5	1.1	0.7	1.1	0.9	1.0	0.7	0.7	0.8	0.9	1.2	1.3
	PrecStd	5.2	1.0	0.9	1.2	1.2	1.2	1.5	1.5	1.3	1.3	1.3	1.2	1.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.8	49.9	67.4	74.5	83.4	87.3	88.0	87.1	85.0	76.0	61.4	52.9
		MCWB	49.5	46.1	59.3	61.8	67.1	73.3	74.2	72.4	70.6	64.2	55.1	50.9
	2%	DB	43.0	42.1	56.8	67.3	78.9	84.0	84.7	82.5	80.1	70.0	57.0	46.3
		MCWB	41.9	40.0	50.4	56.8	65.5	70.8	71.8	70.4	68.2	61.2	53.6	44.3
	5%	DB	37.9	37.6	50.2	62.2	74.3	80.4	81.7	79.6	76.4	65.2	53.2	41.7
		MCWB	36.7	35.7	46.3	53.5	62.6	70.2	70.7	69.2	66.4	59.1	49.8	40.2
	10%	DB	34.3	34.2	44.2	56.9	69.6	76.6	78.8	77.0	72.4	60.9	49.1	37.4
		MCWB	33.3	32.7	40.6	49.4	59.4	67.2	68.5	67.7	64.2	56.1	46.2	36.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.4	47.2	60.2	64.7	70.9	79.3	79.1	76.4	73.4	67.5	57.6	50.8
		MCDB	49.6	49.3	67.3	71.6	79.6	83.3	83.9	81.4	80.7	72.4	59.1	52.9
	2%	WB	42.0	40.2	52.0	59.1	67.4	74.0	75.3	73.4	70.6	63.2	54.2	44.9
		MCDB	42.6	41.7	55.2	65.1	76.0	80.0	80.6	79.4	77.5	67.0	56.7	45.9
	5%	WB	36.9	35.7	46.2	55.0	64.7	71.4	72.9	71.3	68.3	59.8	50.1	40.1
		MCDB	37.8	37.3	50.0	59.9	72.3	78.2	78.8	77.0	73.4	64.6	52.8	41.6
	10%	WB	33.3	32.9	40.8	50.4	61.3	68.8	70.7	69.5	65.8	56.6	46.4	36.1
		MCDB	34.1	34.1	44.1	55.8	67.4	74.4	76.3	74.5	70.9	60.4	49.1	37.1
(35) Mean Daily Temperature Range	MDBR		13.4	14.6	15.4	18.0	19.9	19.0	19.8	19.8	20.3	16.7	12.9	11.0
	5% DB	MCDBR	14.3	16.6	22.9	27.2	25.7	23.3	22.9	22.9	25.0	23.5	19.6	14.2
		MCWBR	14.0	14.7	17.4	18.6	14.8	13.5	12.5	12.7	14.4	15.5	15.9	13.2
	5% WB	MCDBR	14.0	15.6	22.3	24.1	23.0	21.0	19.6	19.5	20.8	21.2	17.9	14.0
		MCWBR	13.9	14.5	18.1	19.4	14.8	14.4	13.0	12.6	13.4	15.2	15.5	13.4
(40) Clear-Sky Solar Irradiance	taub		0.283	0.326	0.322	0.357	0.382	0.412	0.404	0.394	0.363	0.347	0.304	0.283
	taud		2.361	2.191	2.389	2.342	2.316	2.247	2.288	2.305	2.387	2.420	2.499	2.425
	Ebn at Noon		269	271	292	288	282	273	273	273	273	263	261	259
	Edh at Noon		25	35	33	38	40	43	41	39	33	28	22	22
(44) All-Sky Solar Radiation	RadAvg		437	711	1114	1412	1706	1882	1875	1631	1304	777	475	317
	RadStd		56	89	114	157	165	127	125	83	125	75	66	42

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.92	N/A	N/A	N/A	-2.16	-2.74	-223	-322	N/A	N/A
(47) Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page